

Module 08: Lab 08

Exercises

1. Why is it good practice to center and scale before applying k-means clustering?

It is important to center and scale because variables with larger magnitudes may have more influence on the clusters because they weigh in more on the distance calculation. The centering allows comparison between variables and is easier to interpret since the mean is 0.

2. Print the cluster sizes and centers to the R console. Include a screenshot of the output.

```
## {r}
print(km_clusters$size)
print(km_clusters$centers)
```

```
[1] 1275 3189 1312 761

1  1.47753036 -0.3013892
2 -0.48742913 -0.6139217
3 -0.20838565  1.2655088
4 -0.07363698  0.8958214

1 -0.07442448
2 -0.39902544
3 -0.18483146
4  2.11548256
```

3. Use the aggregate() function to calculate the mean of each variable within each cluster. Include a screenshot of the output.

Description: df [4 x 41]

cluster <int>	V1 <dbl>	V2 <dbl>	V3 <dbl>	V4 <dbl>	V5 <dbl>	V6 <dbl>	V7 <dbl>	V8 <dbl>	V9 <dbl>	V10 <dbl>
1	0.47089646	0.012303488	-0.004037244	1.47753036	0.9315513	-0.3013892	-0.04132837	0.0600539	0.07120574	-0.07442448
2	-0.40051534	0.008381724	0.167435602	-0.48742913	-0.2299512	-0.6139217	-0.06166960	-0.1341658	-0.04818380	-0.39902544
3	0.48652362	0.074649497	-0.217773038	-0.20838565	-0.2248895	1.2655088	0.05203055	0.1991881	0.09497949	-0.18483146
4	0.05063266	-0.184436799	-0.319430253	-0.07363698	-0.2094065	0.8958214	0.23796838	0.1182014	-0.08113307	2.11548256

4 rows | 1-11 of 41 columns

4. Assign the observations in `test_features` to the clusters found using the sample of the observations in `training_features` above. Include a screenshot of the output. Hint: a function in the 'clue' R library can be used for this.

Class ids:

```
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[ reached getOption("max.print") -- omitted 1179 entries ]
```